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PROJECT REPORT

PROJECT TITLE:

AI-POWERED PERSONALIZED PASSWORD GENERATOR WITH VISUAL MEMORY SYSTEM (MindLock)

PREPARED BY:

NAME	MATRICS NUM
ALWALEED HASSAN ELYAS HASSAN	B032310752

PREPARED FOR:

MUHAMMAD RAIHAAN BIN KAMARUDIN

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ABSTRACT

Passwords remain the most widely used authentication mechanism for protecting digital resources in modern computing environments. However, the increasing requirement for complex and unique passwords has created significant cognitive burden for users, leading to insecure practices such as password reuse and weak credential selection. Traditional password managers often rely on random password generation and a single master password, which introduces usability challenges and security risks.

An intelligent password management system named MindLock is proposed by this project, which integrates artificial intelligence, cognitive personalization, and multi-factor authentication to enhance both security and usability. MindLock generates personalized and memorable passwords by applying rule-based artificial intelligence techniques on user preferences and cognitive patterns. To strengthen access control, the system incorporates biometric facial verification and visual memory anchors as additional authentication factors.

The system was developed as a web-based application using the FastAPI framework, PostgreSQL for secure data storage, and OpenCV for biometric verification. An AI support agent was also implemented to provide intelligent user guidance through natural language interaction. The performance of the biometric module was evaluated through accuracy testing under different lighting conditions.

The results demonstrate that MindLock is capable of generating secure and memorable passwords while providing reliable authentication and user-friendly interaction. The proposed system shows the potential of intelligent and human-centred approaches in improving modern password management practices.

ABSTRAK

Kata laluan masih merupakan mekanisme pengesahan yang paling banyak digunakan untuk melindungi sumber digital dalam persekitaran pengkomputeran moden. Walau bagaimanapun, keperluan untuk menggunakan kata laluan yang unik dan kompleks telah meningkatkan beban kognitif pengguna, sekali gus mendorong kepada amalan tidak selamat seperti penggunaan semula kata laluan dan pemilihan kata laluan yang lemah. Selain itu, pengurus kata laluan tradisional sering bergantung kepada penjanaan kata laluan rawak dan satu kata laluan utama yang boleh menimbulkan risiko keselamatan dan masalah kebolehgunaan.

Projek ini mencadangkan satu sistem pengurusan kata laluan pintar yang dinamakan MindLock yang menggabungkan kecerdasan buatan, pemperibadian kognitif dan pengesahan berbilang faktor untuk meningkatkan tahap keselamatan dan kebolehgunaan. MindLock menjana kata laluan yang diperibadikan dan mudah diingati dengan menggunakan teknik kecerdasan buatan berasaskan peraturan berdasarkan keutamaan dan corak kognitif pengguna. Bagi mengukuhkan kawalan akses, sistem ini turut mengintegrasikan pengesahan wajah biometrik dan penanda ingatan visual sebagai faktor pengesahan tambahan.

Sistem ini dibangunkan sebagai aplikasi berasaskan web menggunakan rangka kerja FastAPI, PostgreSQL untuk penyimpanan data yang selamat, dan OpenCV untuk pengesahan biometrik. Selain itu, ejen sokongan pintar turut dibangunkan bagi membantu pengguna melalui interaksi bahasa semula jadi. Prestasi modul biometrik dinilai melalui ujian ketepatan di bawah keadaan pencahayaan yang berbeza.

Hasil kajian menunjukkan bahawa MindLock mampu menjana kata laluan yang selamat dan mudah diingati sambil menyediakan pengesahan yang boleh dipercayai serta interaksi yang mesra pengguna. Sistem yang dicadangkan ini membuktikan potensi pendekatan pintar dan berpusatkan manusia dalam meningkatkan amalan pengurusan kata laluan moden.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Password remains the most widely used authentication method for protecting digital resources in this current modern computing environment. Despite major developments in authentication methods such as biometric verification, token-based authentication, and smart security devices, the passwords remain as the primary security layer for the most web applications and online services. This is mostly because of its simplicity, minimal implementation cost, and universal compatibility across different platforms and systems.

However, the increasing demand for unique and complicated passwords has significantly increased the cognitive burden on users. Individual sometimes required to handle dozens of login credentials for a variety of apps such as social medial, bank platform, academic system and corporate services. As the result, many of the users adopt insecure practices such as reuse the passwords, predictable password patterns or storing the passwords in unsafe locations which exposes systems to serious cybersecurity risks.

To address these challenges, an intelligent password generator system is proposed as MindLock that integrates artificial intelligence, visual memory techniques, and multi-factor authentication to enhance both security and the usability. Unlike the traditional password managers that create random strings, MindLock prioritizes user cognition by creating unique passwords based on individual preferences and cognitive tendencies. These passwords are intended to be strong and memorable, with high security standards maintained.

Moreover, MindLock is also provides additional layer of security with multi-factor authentication such as facial verification. By combining human-centred design with intelligent security methods, a practical and scalable solution to modern password

management challenges in a digitally connected society is aimed to be provided by MindLock.

1.2 Problem Statement

Several important issues were motivated the development of MindLock:

i. The average user has over 100 passwords to manage which worsen the situation by causing severe cognitive fatigue and leads to the recycling of weak credentials.
ii. Most password management technologies generate and save random strings of characters which makes the user depend on the application to create memory dependence in turn reducing trust and ownership of the overall security.
iii. Current password storage solutions often rely on a single master for retrieval which creates one critical single point failure without having a multi-factor verification to authorize the retrieval

1.3 Objectives

The objectives of this project are:

i. To design and develop an AI-powered personalized password generator capable of producing strong and secure passwords based on user preferences and cognitive patterns for new application registrations.
ii. To develop a secure web-based database system for storing generated passwords with strong data integrity and availability mechanisms.
iii. To implement a robust multi-factor authentication system to control access to stored passwords.
iv. To provide users with a personalized password management system that assists in securely remembering and retrieving passwords through an intuitive web interface.

1.4 Project Scope

The scope of the MindLock system focuses on the development of an AI-powered, web-based password management application which designed to enhance the password security while improving user memorability. The project's scope is limited to providing the core functionality required to provide personalized password generation, secure storage, and controlled access.

1.4.1 Modules to be developed

The following modules are included within the scope of the project:

i. Personal Preferences Profiler Module

   This module collects the user preferences through a structured survey to identify cognitive patterns for the personalized password generation.

ii. Cognitive Password Generator Module

   This module applies rule-based artificial intelligence techniques to transform user preferences into strong, unique and memorable passwords.

iii. Biometric Verification Module

   This module performs facial verification as a biometric authentication factor to enhance the security.

iv. Visual Memory Assistant Module

   This module provides visual cues using selectable icons to assist users in recalling their passwords.

v. Secure Database and Vault Module

   This module stores the generated passwords securely in a database with controlled access.

vi. AI Support Agent Module

   This module offers an intelligent chatbot interface which assists users with system navigation and account recovery support.

1.4.2 Target Users

The MindLock system is intended for the following target user group:

i. General users who manage multiple online accounts
ii. Students who frequently access academic and online platforms
iii. Organization or individuals who seek improved password security solutions

1.5 Software and Hardware Requirement

The software and hardware requirements used to develop and implement the MindLock system are summarized in the table below.

Table 1.1 Software and Hardware Requirement

Category	Component	Specification / Version
Software	Operating System	Windows 11
	Programming Language	Python 3.10.11
	Frontend Language	HTML
	Styling Engine	CSS
	Client-Side Logic	JavaScript
	Web Framework	FastAPI
	AI Vision Library	OpenCV
	Database Management	PostgreSQL
	Code Editor	Visual Studio Code
Hardware	Processor	Intel Core 5 / AMD Ryzen 5
	Memory (RAM)	8 GB (Minimum)
	Camera	720p HD Integrated Webcam
	Storage	512 GB SSD

1.6 Project Significance

This project is aimed to addressing common cybersecurity issues related to weak and reusable passwords. As multiple online accounts are required to be managed by users, a single master password is relied upon by many existing password management systems which creates a significant security risk. MindLock is a web-based, AI-driven password management system designed to balance security and usability. Personalized passwords are generated by the system based on user preferences and cognitive patterns which helps complex passwords to be remembered by users while strong security is maintained.

This strategy reduces the dependency on random password generators while encourages more involvement with the personal security methods. In addition, multi-factor authentication such as biometric face verification and visual memory cues is included in MindLock to reduce the risk of unauthorized access. The project demonstrates how a real-world cybersecurity problems can be addressed by intelligent system while a user-friendly and accessible web platform is maintained.

1.7 Summary

This chapter provided an overview of the MindLock project by discussing the background and motivation behind the development of an intelligent password generator system. The key cybersecurity challenges related to password management were identified, leading to the formulation of clear problem statements. In addition, the project objectives and scope were created to define the system's expected functions and constraints. The chapter also highlighted the MindLock's importance in tackling real-world security concerns using a human centered and AI-driven strategy. Overall, this chapter set a strong foundation for the subsequent chapters which further detail the methodology, system design, implementation, and evaluation of the proposed system.

CHAPTER 2: METHODOLOGY

2.1 Intelligent System Development Methodology

The MindLock system was developed using a planned and systematic intelligent system development process which ensures that the project is organized and effective. This approach follows to the Workshop II guidelines and provides an organized path from problem identification to system implementation and evaluation.

The methodology adopted in this project consists of several key phases which is the requirement analysis, system design, implementation, testing, and evaluation. Each phase has an importance to ensure that the intended objectives are satisfied by the finished system while also addressing the identified cybersecurity issues.

During the requirement analysis phase, the project scope, objectives, and functional requirements were identified based on frequent password management issues that encountered by users. This phase includes examining the existing password generator systems, identifying their weakness, and deciding the features needed to increase security and usability.

The system design phase focused on designing the overall architecture of MindLock, including module interactions, data flow, and database structure. At this stage, the system was divided into multiple functional modules such as user preference profiling, cognitive password generation, biometric verification, visual memory assistance, secure password vault and AI support assistance. The design phase ensures that each module can function independently while remaining integrated within the overall system.

The implementation phase next required developing the system with the proper programming languages, frameworks, and tools. MindLock was built as a web application with FastAPI and Python for backend processing, PostgreSQL for the database administration, and the OpenCV for facial verification. The effective

password generating logic was created with a rule-based AI technique that converts all the user choices into safe and memorable passwords.

The testing phase was conducted to verify that each module functions correctly and that the system operates as intended. Functional testing, integration testing, and basic user acceptance testing were performed to identify errors, improve performance, and ensure system reliability.

Finally, the evaluation phase determined if the developed system had effectively met its objectives. This step included testing the effectiveness of the AI-generated passwords, the accuracy of biometric verification, and the system's general usefulness from a user perspective.

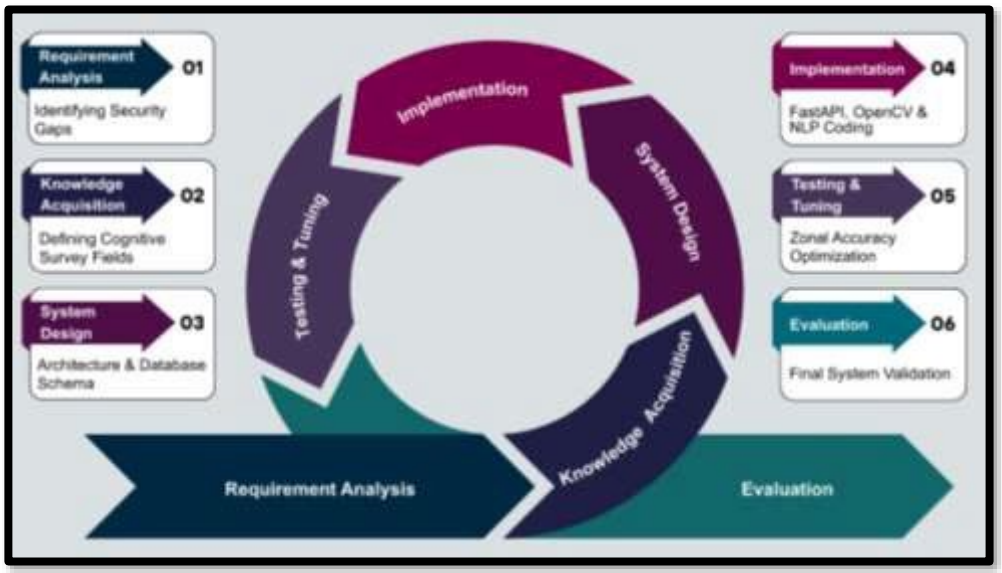

Figure 2.1 Agile Development Lifecycle

2.2 Intelligent System Development Technique

MindLock applies several intelligent system development techniques to address the limitations of the traditional password management systems. Artificial intelligence, cognitive modelling, biometric authentication, and visual memory assistance are combined by these techniques to enhance the system security and user experience.

One of the primary techniques used in MindLock is rule-based artificial intelligence for password generation. User preferences collected through a structured

survey are processed using predefined transformation rules such as character substitution, pattern mixing, symbol insertion, and length control. This technique enables the generation of passwords that are both strong and meaningful to the user, improving memorability without compromising security.

Another important technique used is cognitive-based personalization. Instead of creating completely random passwords, the system uses personal information such as interests, favourite things, and nicknames to generate passwords that correspond to the user's cognitive memory patterns. This human-centred approach minimizes cognitive fatigue while increasing user involvement with security measures. Biometric verification is used for authentication by detecting and verifying face features. OpenCV is used to recognize and compare facial features during the registration and login procedure. By this verification method, an additional layer of protection is provided by this biometric technology by guaranteeing that only the authorized users have access to their saved passwords.

Additionally, MindLock includes a visual memory support technique using the visual icons. These symbols function as memory anchors which allows users to link individual passwords with visual clues. This method improves recall while reducing reliance on written or stored passwords. In addition, an AI support agent module is also included to help users with system navigation, password generating, and account recovery. The support agent provides suitable replies based on specified conversational rules, further improving system usability and satisfaction with it.

2.3 Summary

This chapter defined the methodology and intelligence strategies used when developing the MindLock system. The intelligent system development methodology provides a structured approach that guides the project through requirement analysis and system evaluation. In addition, the intelligent development strategies used in MindLock such as rule-based AI, cognitive preferences, biometric verification, and visual memory assistance were thoroughly discussed. These strategies allow MindLock to work as a safe, user-centred, and intelligent password management application. The following chapter discusses at the system requirements and existing password management solution.

CHAPTER 3: ANALYSIS

3.1 Analysis of Current Application

In the current digital environment, password management remains a major challenge for users across various platforms. Most existing applications are relied on heavily by traditional authentication methods, primarily using usernames and passwords as the sole means of access control. While many systems are enforced password complexity rules, such as minimum length and character variety, these measures often fail to consider the cognitive limitations of users.

Conventional password managers generate random sequences of characters and safely store them in a centralized vault. Although this strategy strengthens passwords, it greatly reduces memorability. As a result, users become too dependent on the password manager, losing familiarity and control of their credentials. In many cases, weak passwords are reused by users or written down by them, which undermines overall system security.

Another major limitation of current applications is the reliance on a single master password. A critical single point of failure is created where the compromise of one credential can expose all stored passwords. Furthermore, many systems lack additional verification measures like biometric identification or contextual validation, leaving them open to unwanted access.

In terms of usability, current systems frequently prioritize security enforcement over user experience. Trust and adoption are reduced due to a lack of personalized cognitive assistance, and user engagement. These flaws point to the need for a more intelligent, user-centred password management system that combines security needs with human memory and behaviour.

3.1.1 User Use Case Description

This section includes detailed case descriptions to help users understand how to interact with the MindLock system. These use cases explain the primary user activities and system reactions, therefore clarifying functional requirements and system behavior from the user's perspective. Each use case describes the actor involved, the pre-conditions, the primary order of events, and the intended outcome following successful execution.

Table 3.1 Use Case 1 - Registering with Cognitive Survey

Item	Description
Use Case Title	Registering with AI Survey
Actor	New User
Pre-condition	The user wants to access to the MindLock web application and is not yet registered in the system.
Main Flow	**Step 1:** The user accesses the registration page through the MindLock interface. **Step 2:** The user enters basic account information such as username and email. **Step 3:** The system presents a structured cognitive survey requesting personal preference data (e.g., nickname, hobby, favourite number, favourite place). **Step 4:** The user completes and submits the survey form. **Step 5:** The system stores the user details and preference data securely in the database and prepares the data for password generation.
Post-condition	The user account is successfully created and personalized cognitive data is stored for future AI-based password generation.

Table 3.2 Use Case 2 - Accessing the Password Vault

Item	Description
Use Case Title	Accessing Vault through Biometric Verification
Actor	Registered User
Pre-condition	The user is registered in the system, has completed facial enrolment, and has previously stored passwords in the vault.
Main Flow	**Step 1:** The user logs in using basic credentials (username and password). **Step 2:** The system activates the webcam and requests facial verification. **Step 3:** The biometric module captures a live facial image and performs face detection using OpenCV. **Step 4:** The system compares the live image with stored facial data using zonal histogram analysis. **Step 5:** If the Zonal Accuracy Score exceeds the predefined threshold, access is granted to the password vault.
Post-condition	The user gains secure access to the password vault and can view or retrieve stored credentials.

These use case descriptions clearly demonstrate how MindLock combines usability and security through intelligent system behaviour. The registration use case is highlighted by how cognitive data is collected to support personalized password generation, while the vault access use case is emphasized by the role of biometric verification in enforcing multi-factor authentication. Together, these interactions reflect the system's human-centred and security-driven design approach.

3.2 Analysis of Proposed Intelligent System

The proposed system, MindLock, is designed to overcome the limitations identified in existing password management applications. Unlike traditional systems, MindLock adopts an intelligent and human-centred approach by incorporating artificial intelligence, cognitive personalization, and multi-factor authentication.

User preferences and personal attributes are analyzed by MindLock to generate passwords that are both secure and memorable. Instead of producing random character sequences, rule-based AI techniques are applied by the system to transform the meaningful user data into strong passwords. Cognitive fatigue is reduced by this approach and better password practices are encouraged.

From a security perspective, the proposed system introduces multi-factor authentication, combining knowledge-based authentication, biometric verification, and visual memory cues. This layered security model significantly reduces the risk of unauthorized access, even if one authentication factor is compromised.

The system also improves usability by offering a structured and interactive web interface. Users are guided through registration, password generation, and retrieval processes with the support of an AI-based support agent. This intelligent assistance helps users understand system features and improves overall interaction.

By integrating intelligence into both security and usability components, MindLock provides a comprehensive solution that aligns with modern cybersecurity requirements while remaining accessible to general users.

Table 3.3 Functional Requirements Specification

ID	Requirement Name	Description
FR-01	Data Capture	System must collect 12 survey points
FR-02	AI Generation	System must produce 10 Leetspeak variants
FR-03	Biometric Check	System must verify face before vault access
FR-04	AI Assistance	Chatbot must respond to security queries

Table 3.3 summarizes the main functional requirements of the MindLock system which describes the important features needed to support personalized password generation, secure authentication and intelligent user assistance.

The first requirement defined the collection of 12 cognitive and preference inputs from users during registration, which serve as the foundation for password generation. The second requirement specifies that the AI module must generate multiple secure password variants using rule-based transformation. The third requirement enforces biometric verification before allowing access to the password vault, providing an additional layer of protection. The final requirement highlights the role of the AI support agent in responding to user queries and assisting with system usage.

Table 3.4 Non-Functional Requirements Specification

ID	Requirement Name	Description
NFR-01	Security	The system must encrypt all stored passwords and enforce biometric verification before access.
NFR-02	Performance	The system should generate passwords within 2 seconds of user request.
NFR-03	Reliability	The system should maintain at least 95% uptime during normal operation.
NFR-04	Usability	The user interface must be simple and easy to use for non-technical users.
NFR-05	Scalability	The system must support multiple users and allow future module expansion.

Table 3.4 presents the non-functional requirements defined for the MindLock system. These requirements describe the quality attributes and operational constraints that ensure the system performs effectively, securely, and reliably beyond its core functional features.

The first requirement emphasizes security, where all stored passwords must be encrypted and biometric verification must be enforced before vault access. This

ensures that sensitive user data is protected and that unauthorized access is prevented even if login credentials are compromised.

The second requirement focuses on performance, specifying that password generation should be completed within two seconds after a user request. This ensures that the system remains responsive and provides a smooth user experience without noticeable delays.

The third requirement addresses reliability, requiring the system to maintain at least 95% uptime during normal operation. This guarantees that the system remains accessible to users and minimizes service interruptions.

The fourth requirement highlights usability, stating that the user interface should be simple and intuitive, especially for non-technical users. This supports ease of adoption and encourages proper use of the system's security features.

Finally, the scalability requirement ensures that the system can support multiple users and allow future expansion of modules and features. This is important for long-term deployment, particularly in organizational environments where the number of users may increase over time.

3.3 Structure Chart of Proposed Intelligent System

The structure chart of MindLock illustrates the hierarchical organization of the main system components and their functional relationships. At the highest level, the MindLock system is divided into three core subsystems: User Management, Intelligence Module, and Vault Management, which together provide the complete password management functionality.

The User Management component handles user-related operations including registration, survey data collection, login, and session management. This module acts as the primary interaction point between the user and the system, ensuring secure account creation and authentication.

The Intelligence Module contains the core intelligent components of the system. It includes the AI Password Generator which produces personalized passwords, Biometric Verification module for facial authentication, and the AI Support

Agent, which assists users with system navigation and security queries. This module is responsible for implementing the artificial intelligence and decision-making processes of MindLock.

The Vault Management component manages secure password storage and retrieval. It consists of the Encrypted Storage module which stores user credentials securely, and the Visual Memory Anchors module which provides visual cues to assist users in recalling their passwords.

This hierarchical modular structure improves system clarity, maintainability, and scalability by clearly separating responsibilities among system components while ensuring seamless integration within the overall architecture.

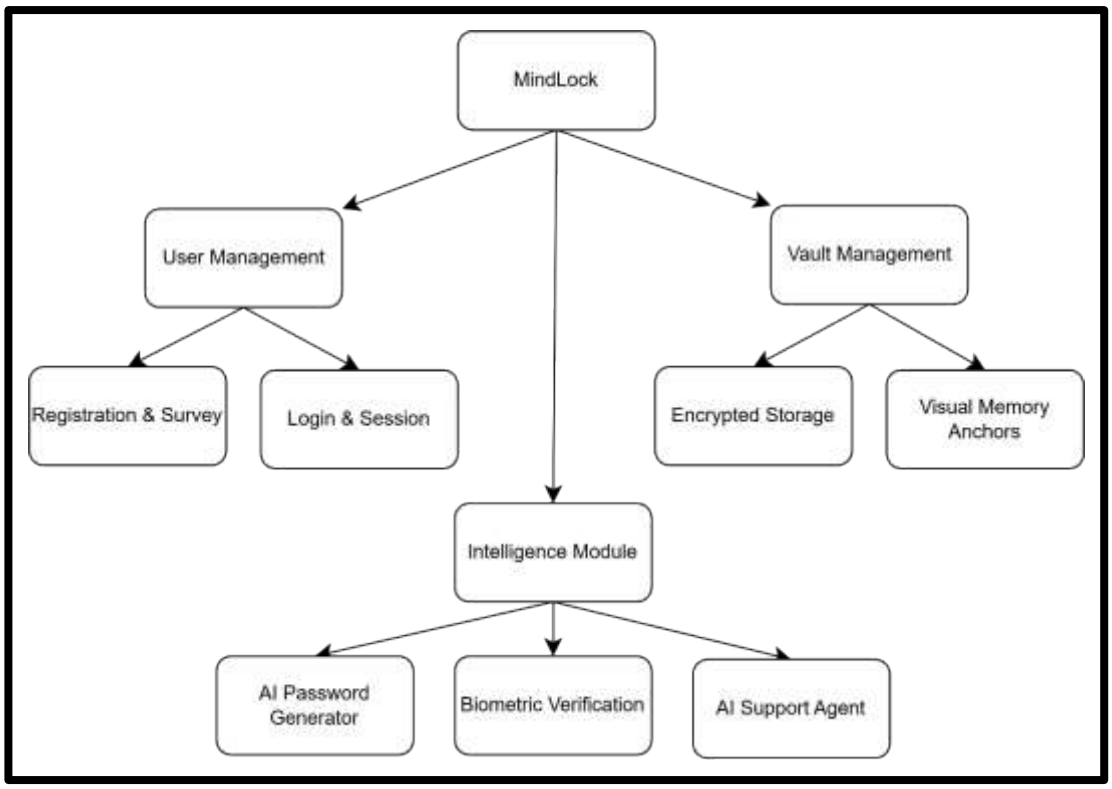

Figure 3.1 Structured Chart

3.4 Work Breakdown

The development of the MindLock system was organized into a structured set of manageable tasks to ensure systematic progress and effective teamwork. The Work Breakdown Structure (WBS) divides the project into several major development phases, each focusing on a specific aspect of system construction.

The project began with the Project Planning and Requirement Analysis phase which included intelligent system selection, proposal preparation, and system flow planning. This phase established the project scope, objectives, and overall development strategy. Next, the User Interface and System Structure phase focused on initial user interface development and system layout design.

The development process then continued with specialized module implementation phases. The Biometric Verification Module phase involved the development and enhancement of the facial recognition component, followed by the Core Intelligent Modules Development phase, which included the cognitive password generator, personal preference profiler integration, and visual memory assistant implementation. The AI Support Agent Module phase focused on designing conversational rules and implementing intelligent response features.

After completing the core modules, the Backend and Frontend Integration phase was carried out to integrate all system components using FastAPI and PostgreSQL. Each module was developed and tested individually before being integrated into the complete system.

This structured work breakdown ensured that all development activities were clearly defined, properly scheduled, and completed within the allocated timeframe, leading to efficient project execution and successful system delivery.

TASK	ASSIGNED TO	START	END
Project Planning & Requirement Analysis			
Intelligent System Selection	All Members	06-10-2025	10-10-2025
Project proposal preparation and submission	All Members	13-10-2025	17-10-2025
System flow planning and storyboard preparation	All Members	20-10-2025	24-10-2025
System flow finalization and task allocation	All Members	27-10-2025	31-10-2025
User Interface & System Structure			
Initial UI development	Guo Heng	03-11-2025	07-11-2025
Biometric Verification Module			
Facial recognition module development	Keerthana	10-11-2025	14-11-2025
Facial recognition enhancement and accuracy testing	Alwaleed	17-11-2025	21-11-2025
Core Intelligent Modules Development			
Cognitive password generator	Guo Heng	10-11-2025	21-11-2025
Personal preference profiler integration	Guo Heng	10-11-2025	21-11-2025
Visual memory assistant	Danish Farhan	10-11-2025	21-11-2025
AI Support Agent Module			
Conversational Rule Design	Alwale & Danish	01-12-2025	05-12-2025
Feature Explanation Responses	Alwaleed & Danish	01-12-2025	05-12-2025
Backend & Frontend Integration			
Full system integration using FastAPI and PostgreSQL	All Members	08-12-2025	19-12-2025

Figure 3.2 Work Breakdown Structure

3.5 Summary

The limitations of existing password management applications were examined by this chapter, highlighting key issues related to security weakness, usability challenges, and over-reliance on traditional authentication mechanisms. These findings highlighted the importance of a more intelligent and user-centred password management system. The proposed MindLock system was then analyzed with respect to its security mechanisms, usability improvements, and intelligent features that address the identified problems. In addition, the interaction between system modules within a layered architecture was illustrated by the structured chart, demonstrating clear functional separation and integration. The work breakdown further explained how the development tasks were systematically organized and executed throughout the project. Overall, this analysis establishes a solid foundation for the next chapter which presents the detailed system design of the MindLock Application.

CHAPTER 4: DESIGN

4.1 Introduction

This chapter presents the detailed system design of the MindLock password generator system. It explains how the proposed system is organized and how its components work together to deliver a secure, intelligent, and user-friendly password management experience.

The system design is based on the requirements and analysis discussed in the previous chapter and serves as a guide for the system implementation phase. By defining the system architecture, module structure, and interactions between components, the design ensures that the implementation process is clear, structured, and well-organized.

MindLock is designed using a modular approach to ensure ease of maintenance and future scalability Each module performs a specific function while remaining seamlessly integrated with the overall system workflow. Intelligent features such as personalized password generation, biometric verification, visual memory assistance, and user support are supported by this structure, helping strong security and a positive user experience be achieved by the system.

4.2 Intelligent System Architecture

A layered Model-View-Controller client-server architecture is adopted by MindLock. This separation of concerns is particularly important for security-centric systems, as sensitive data processing and authentication logic are isolated from the user-facing interface.

The Presentation Layer is responsible for user interaction and interface rendering. It is implemented using Jinja2 templates combined with Glassmorphism-

styled CSS to provide a modern and visually intuitive interface. This layer captures user inputs such as survey data and webcams frames, displays visual memory anchors, and uses JavaScript to manage asynchronous requests to backend AI services.

The Application Logic Layer is powered by the FastAPI framework and functions as the core processing unit of the system. This layer contains the rule-based AI password generator, the OpenCV based biometric verification logic, and the Natural Language Processing components used by the AI support agent. It coordinates system workflows, enforces authentication rules, and manages access control.

The Data Layer utilizes a PostgreSQL relational database to store and manage system data. Three primary entities are maintained by it which are user credentials, personalized preferences, and the encrypted password vault. This layer ensures data integrity, consistency, and secure storage.

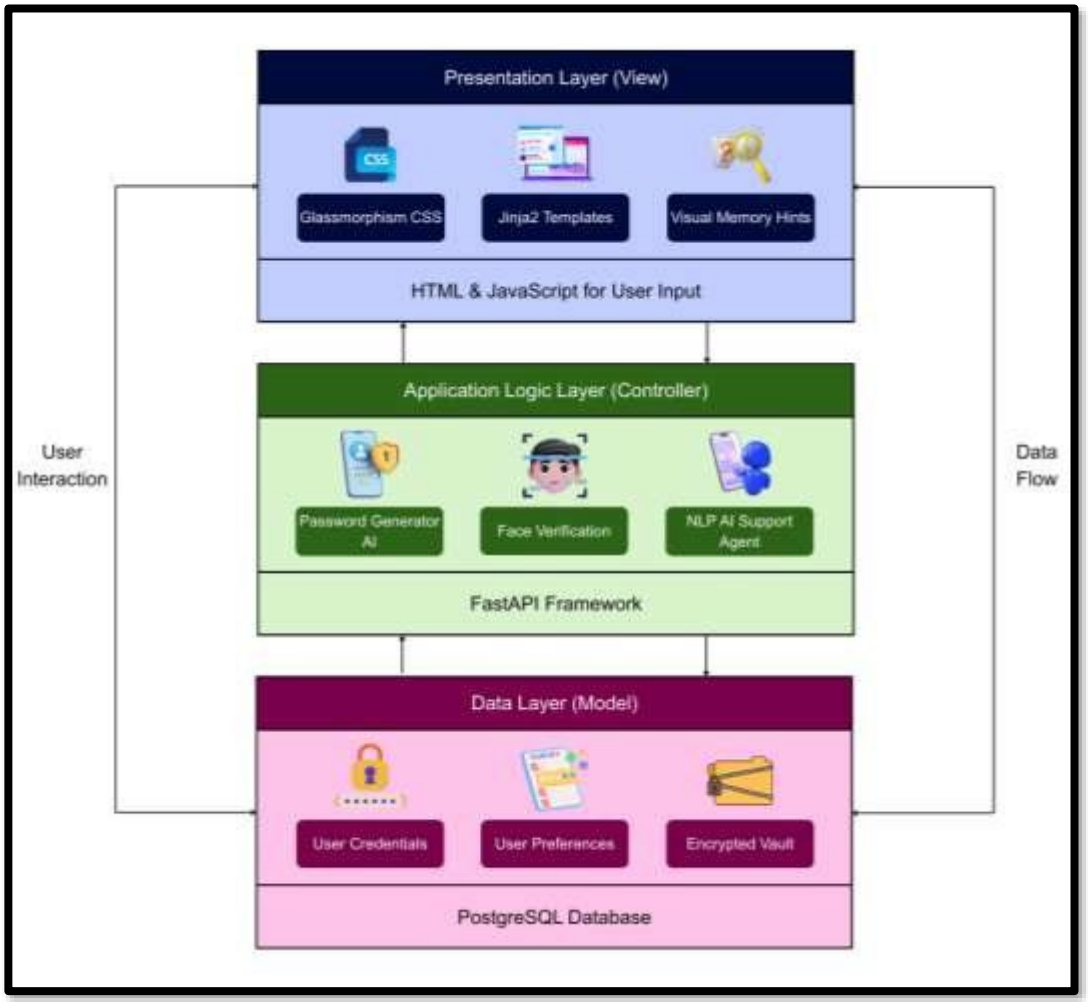

Figure 4.1 MVC Architecture Design

4.3 Module 1: Personal Preference Profiler Module

The Personal Preference Profiler Module serves as the primary data collecting component of the MindLock system. During user registration, individuals complete a structure 12 cognitive survey designed to capture meaningful personal attributes that support memory-based password generation.

The collected data includes several categories:

- Identity indicators: Nickname, childhood nickname
- Interest indicator: Favourite sport, hobby, favourite celebrity
- Numerical anchors: Favourite number
- Locational anchors: Favourite place, favourite café

Rather than storing these inputs for immediate retrieval or search, these inputs are formatted by the module as structured inputs for the Cognitive Password Generator. By ensuring that each password is created from existing user memories, password recall probability is greatly improves by this module.

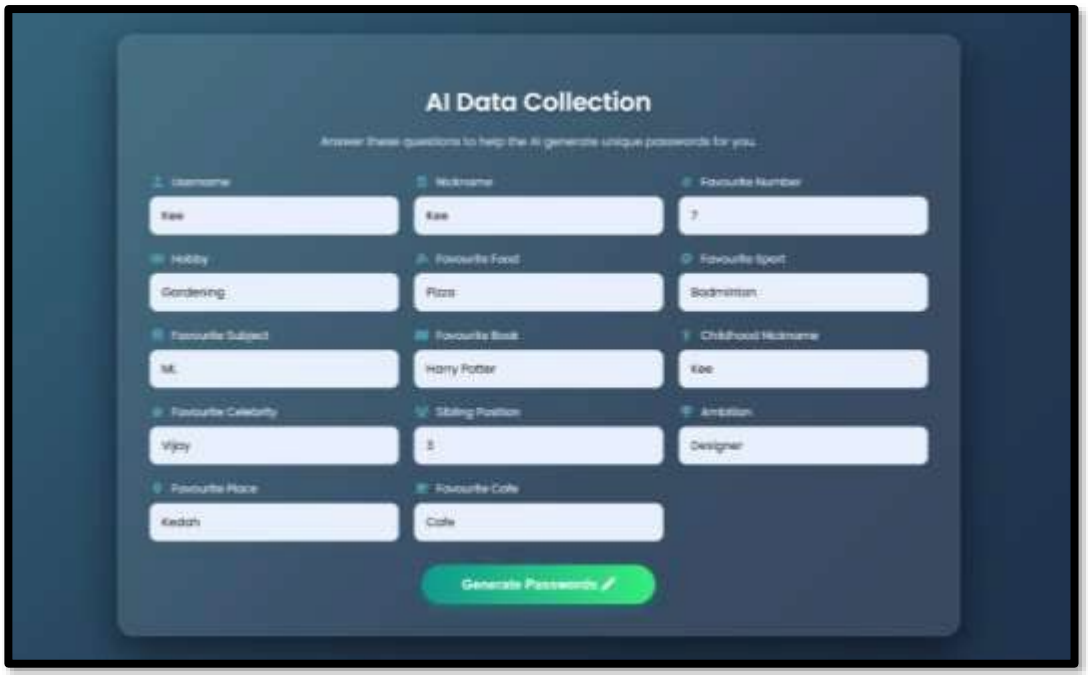

Figure 4.2 Personal Preference Profiler Module

4.4 Module 2: Cognitive Password Generator Module

The Cognitive Password Generator Module is Mindlock's primary artificial intelligence component. This module substitutes the traditional random string generation with a rule-based transformation engine that provides a balance between security and memorability.

The password generation process follows a multi-step transformation pipeline:

- Data selection: Two preference fields are randomly paired such as the nickname and the hobby.
- Leetspeak Substitution: The linguistic transformation rules are applied
- Entropy Injection: A randomized 3-digit string derived from the user's favourite number, along with a special symbol is injected to satisfy complexity requirements such as minimum length, mixed case and symbol usage.

This structures approach confirms that security requirements are fulfilled by passwords while still being cognitively relevant to users.

Table 4.1 Leetspeak Transformation Knowledge Base

Input Char	AI Substitution	Example
A / a	4	Apple → 4ppl3
E / e	3	Eagle → 34gl3
I / i	1	Ice → 1c3
O / o	0	Ocean → 0c34n
S / s	5	Sky → 5ky

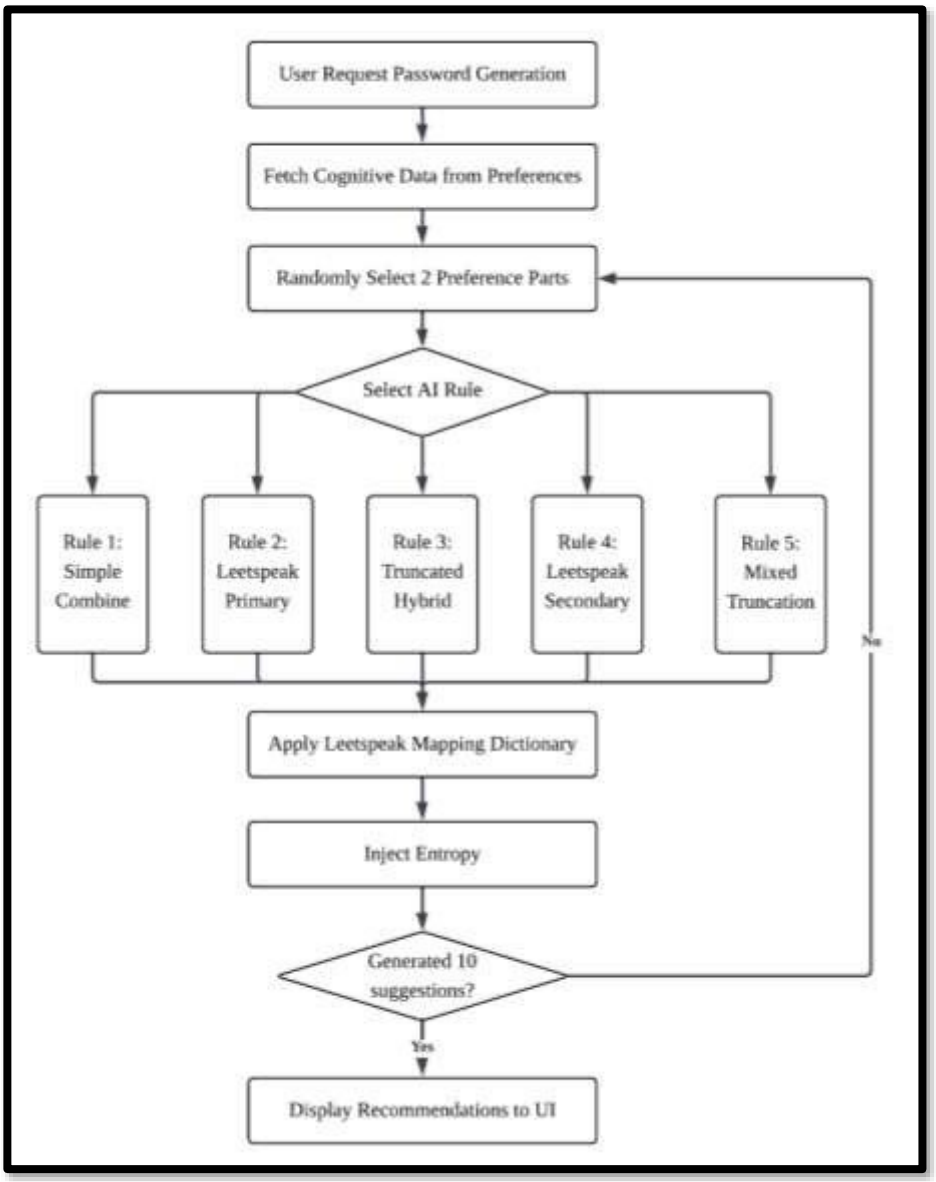

Figure 4.3 Password Generation Process Diagram

4.5 Module 3: Biometric Verification Module

A Computer Vision-based security layer is introduced by the Biometric Verification Module that strengthens system access control. Two distinct phases are operated in by this module. During the detection phase, the human face is identified and localized within the live webcam feed by Haar Cascade classifiers. Once detection is successful, the verification phase is proceeded to by the system where a Zonal Histogram Comparison is performed.

The detected face is divided into three regions which is the forehead, nose, and mouth. Pixel distribution histograms from each zone are compared against stored registration data. A combined Zonal Accuracy Score is computed, and access is granted only if a predefined threshold of 0.3 is exceeded by the score. This ensures that possession of a password alone is insufficient to compromise the password vault.

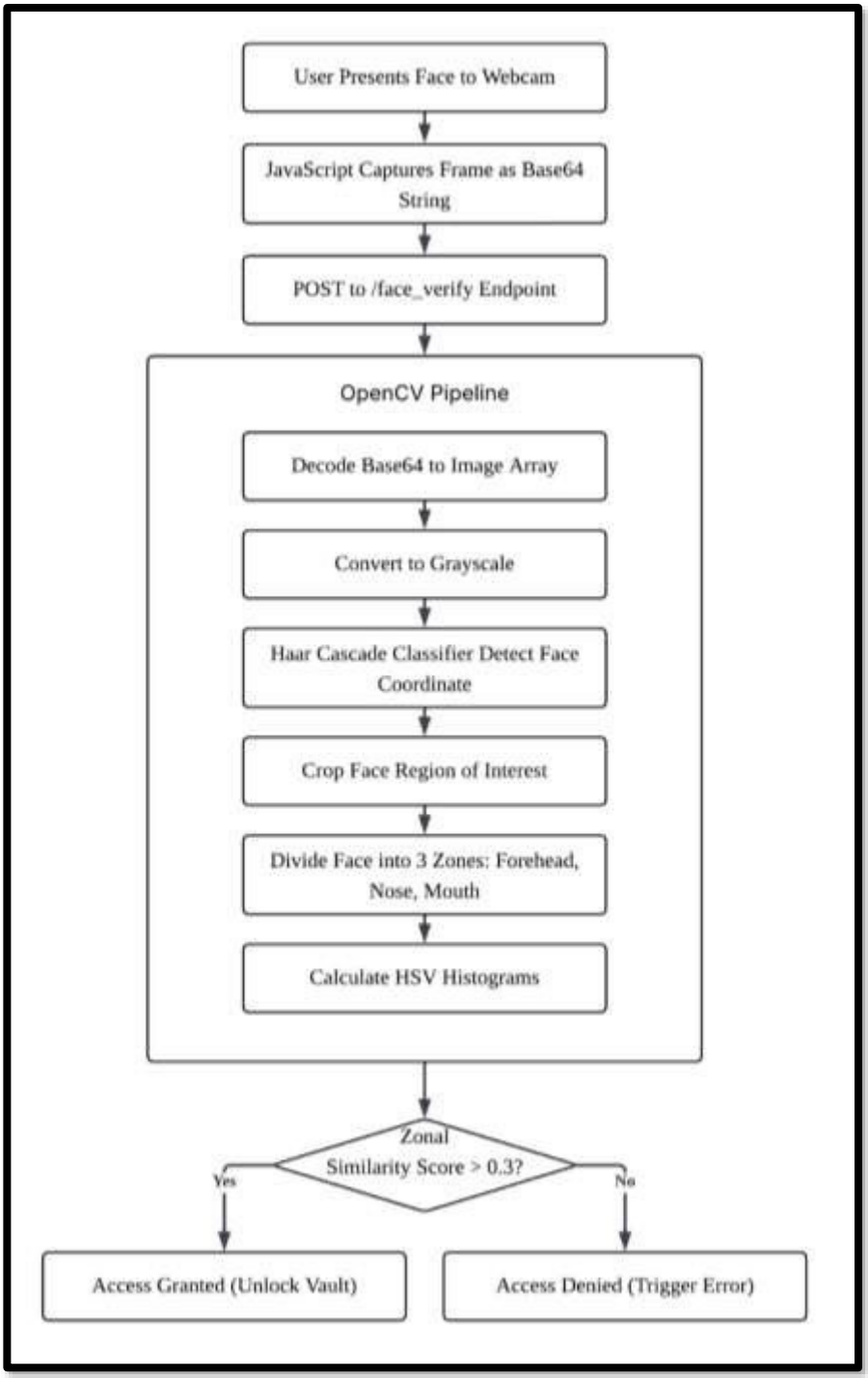

Figure 4.4 Facial Verification Process Diagram

4.6 Module 4: Visual Memory Assistant Module

The Visual Memory Assistant Modules is based on the psychological principle of visual anchoring. When a password is generated by users for a specific platform, such as Instagram, a representative icon is selected by them from a curated set, including examples like mountains, fire or food-related images.

Within the user dashboard, passwords remain visually blurred until interaction, while the selected icon remains visible at all times. This icon is acted upon as the memory anchor which allows complex password strings to be associated by users with simple and easily recognizable visual cues, thereby improving recall without compromising security.

Figure 4.5 Visual Memory Anchors Used for Password Recall

4.7 Module 5: Secure Database and Vault Module

The Secure Database and Vault Module is responsible for maintaining data integrity and confidentiality. All create, read, update, and delete (CRUD) operations related to stored passwords are managed by it.

Data integrity is enforced using key constraints within the PostgreSQL database to ensure that each stored password is always linked to a valid user account. From a security perspective, readable forms of stored passwords are not displayed unless biometric verification is successfully completed by the user. This design decision helps prevent threats such as unauthorized viewing.

Table 4.2 Data Dictionary – User Entity

Field Name	Type	Constraint	Description
id	Integer	PK, Auto Increment	Unique entry ID
email	String	Unique	Login email
hashed_password	String	Not Null	Master key (Encrypted)
face_pickle	Binary	Nullable	Biometric Signature

Table 4.3 Data Dictionary – saved_password Entity

Field Name	Type	Constraint	Description
id	Integer	PK	Unique entry ID
platform	String	Not Null	Platform name
username	String	Not Null	Platform Username
password	String	Not Null	The AI Generated Password
image_hint	String	Not Null	Selected Memory Icon

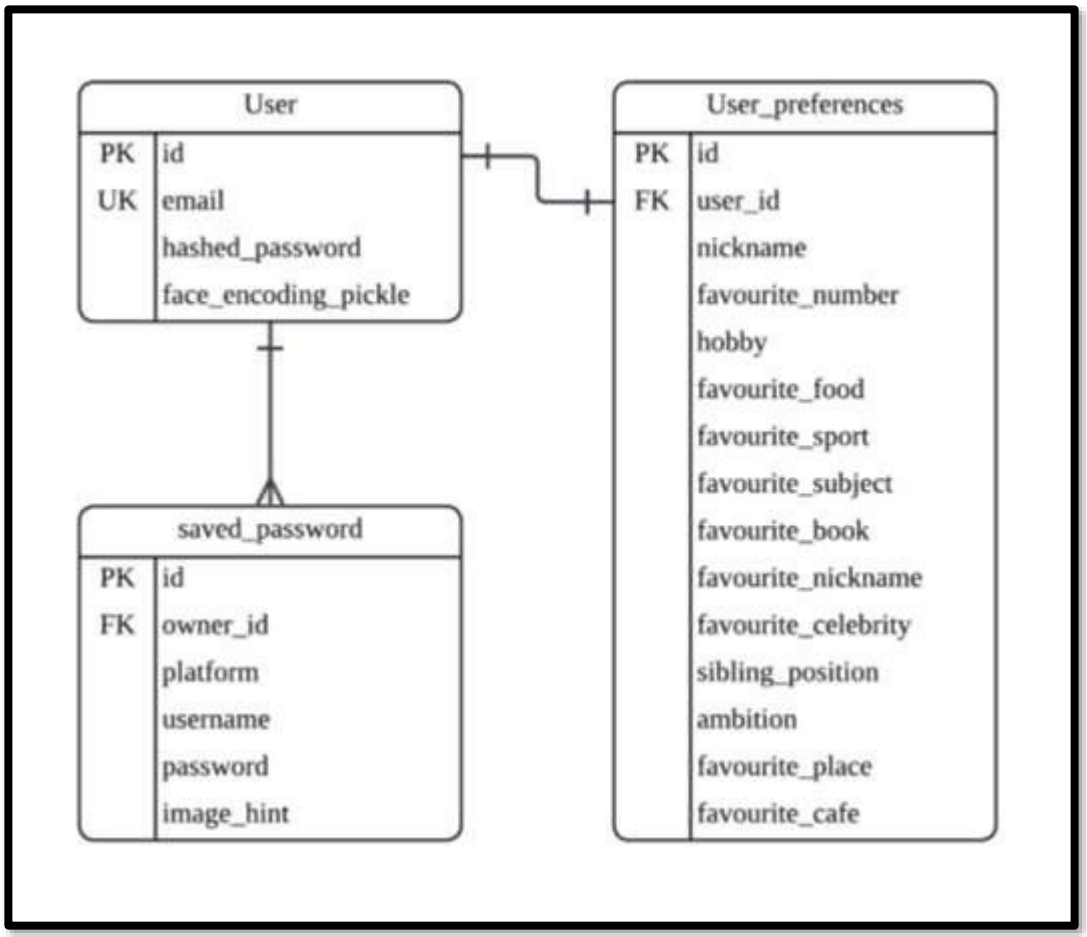

Figure 4.6 Entity-Relationship Diagram

4.8 Module 6: AI Support Agent Module

An interactive Natural Language interface is provided by the AI Support Agent to assist users with system usage and security-related queries. It is designed using intent recognition techniques, allowing questions to be asked by users such as “How do I recover my password?” or “Is my data safe?”. A hybrid approach is adopted by the module:

- Rule-based logic for common navigation and system-related questions.
- Large Language Model (LLM) API integration for complex explanations related to security and system behaviour.

This approach reduces the learning curve for new users and enables continuous, on-demand support.

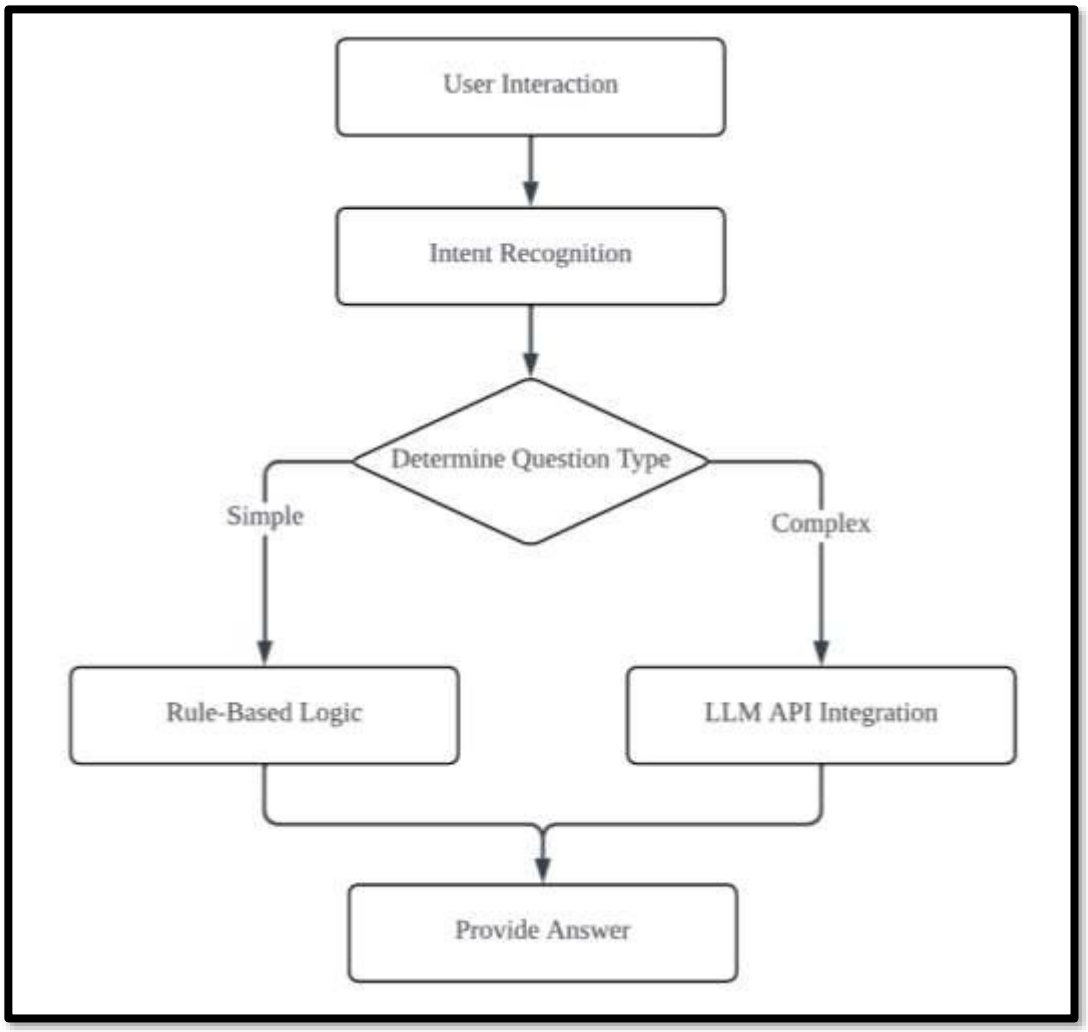

Figure 4.7 AI Support Agent Workflow

4.9 Summary

This chapter presented the detailed system design of the MindLock personalized password generator system. The intelligent system architecture was discussed, followed by a detailed explanation of each functional module and their roles within the overall system. The modular and layered design ensures that the system is secure, scalable, and maintainable while supporting intelligent and user-centred features. In addition, the clear separation of system components allows for easier future enhancements and troubleshooting. This design is served as a strong foundation for the next chapter, which will focus on the implementation details of the MindLock system.

CHAPTER 5: IMPLEMENTATION

5.1 Introduction

This chapter presents the implementation phase of the MindLock intelligent password generator system. The implementation phase acts as a bridge between the system design described in Chapter 4 and a fully functional, deployed software solution. The main objective of this phase was convert the modular architectural designs into executable Python code while ensuring that all intelligent components operate cohesively within single system.

During this phase, particular focus was placed on integrating the FastAPI backend with a responsive, Glassmorphism-styled frontend. The implementation process was focused on ensuring seamless communication between system modules, reliable execution of intelligent features such as biometric verification and rule-based password generation, and secure handling of user data. This chapter is discussed regarding the development environment, configuration management strategy, and how intelligence was embedded into the system through algorithms and API integration.

5.2 Intelligent System Development Environment

The development of the MindLock was carefully selected to support a high-performance, asynchronous web application capable of handling real-time user interaction and intelligent processing. Several technologies were integrated to form the core development stack.

The Fast API framework was chosen as the backend framework due to its support for asynchronous server gateway interface (ASGI). Webcam streams, biometric verification, and AI logic may be processed by the system without blocking the user interface which results in a responsive user experience.

The OpenCV and Google Generative AI are integrated by the intelligence layer of the system. OpenCV was implemented for real-time computer vision tasks, specifically facial detection and verification using Haar Cascade Classifiers. For Natural Language Processing, the MindLock Assistant was powered by the Gemini Pro library which enables intelligent interpretation of user queries and dynamic response generation.

For data persistence, PostgreSQL was selected as the relational database management system. SQLAlchemy was used as the Object-Relational Mapper which allows the database tables to be represented as Python objects. This approach improved code readability. Simplified database operations, and reduced security risks such as SQL injection.

The frontend was developed using Jinja2, HTML, and CSS. Dynamic data rendering from the backend to the user interface was enabled by Jinja2, while a modern "Future-Dark" Glassmorphism aesthetic was implemented using the CSS through backdrop-filter properties.

5.3 Intelligent System Configuration Management

MindLock was created using an organized and modular file structure to assure system stability and compliance with professional software engineering methods. This configuration management technique allows different system components to be designed, tested, and maintained separately.

The system directory structure is organized as follows:

- main.py serves as the central entry point of the application. It defines all API routes for navigation and triggers the execution of intelligent modules.
- models.py defines the database schema using SQLAlchemy classes which includes tables such as User and SavedPassword.
- database.py manages database connectivity by defining the connection string and initializing the database engine.
- crud.py contains all Create, Read, Update, and Delete operations, isolating database queries from application logic.

- The /templates directory stores HTML templates such as dashboard.html and face_verify.html.
- The /static directory contains CSS files and client-side image assets.

This separation of concerns ensures that changes to the database schema in models.py are not required by updates to AI logic in main.py. As a result, the system becomes easier to debug, maintain, and extend in future development phases.

Table 5.1 FastAPI Endpoints and AI Function Mapping

URL Path	HTTP Method	Python Function	Intelligence Module
/register	POST	handle_survey( )	Preference Profiler
/face-verify	POST	verify_face_math( )	Biometric Vision
/api/chat	POST	chat_with_ai( )	AI Chatbot Agent
/generate	GET	rule_engine( )	Cognitive Generator

5.4 Completing Task via Intelligence

The MindLock system's intelligence is divided into multiple intelligent tasks which automate decision making and improve system security, rather than focusing in a single component.

5.4.1 Rule-Based Cognitive Password Generation

MindLock uses a Rule-Based Expert System to generate unique passwords. Unlike traditional password generators that use random ASCII characters, MindLock use a Leetspeak Transformation Algorithm to generate passwords that are both safe and memorable.

The system selects relevant strings from the Personal Preference Profiler and uses a predetermined dictionary-based mapping, such (a: 4, e: 3, i: 1, o: 0, s: 5). The algorithm generates passwords like F00tb4llP1zz4! by randomly choosing two user interests, such as Football and Pizza, then transforming them using leetspeak. This

approach achieves what is referred to as memorable complexity, combining personalization with strong security characteristics.

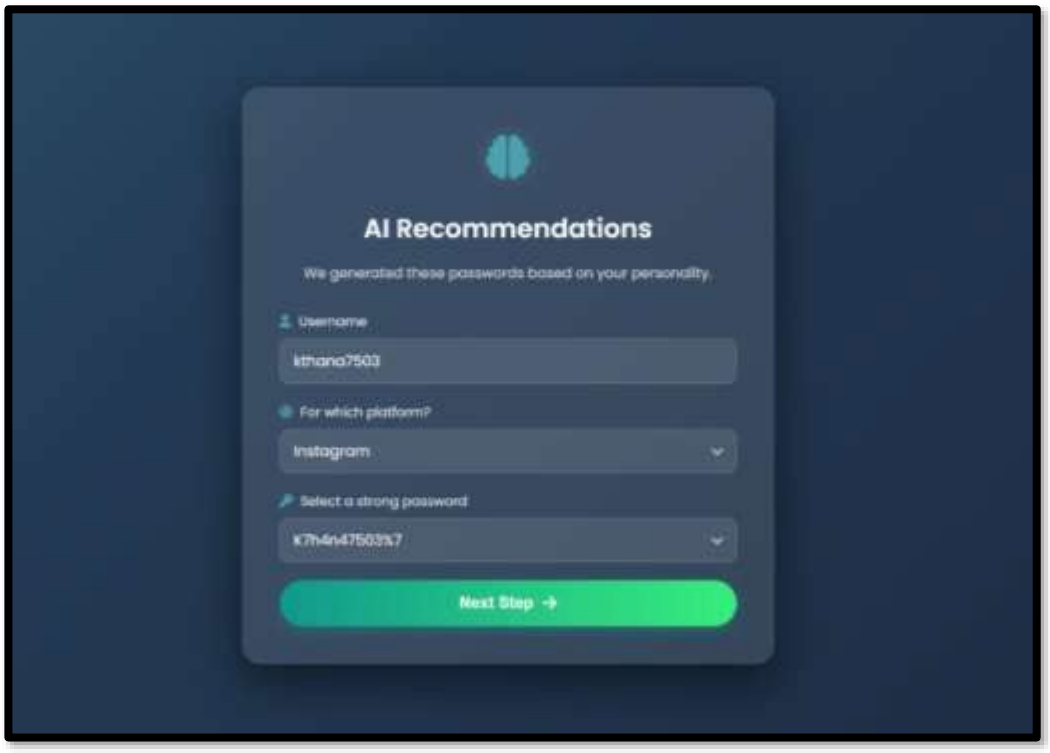

Figure 5.1 Password Generation Based on User Preferences

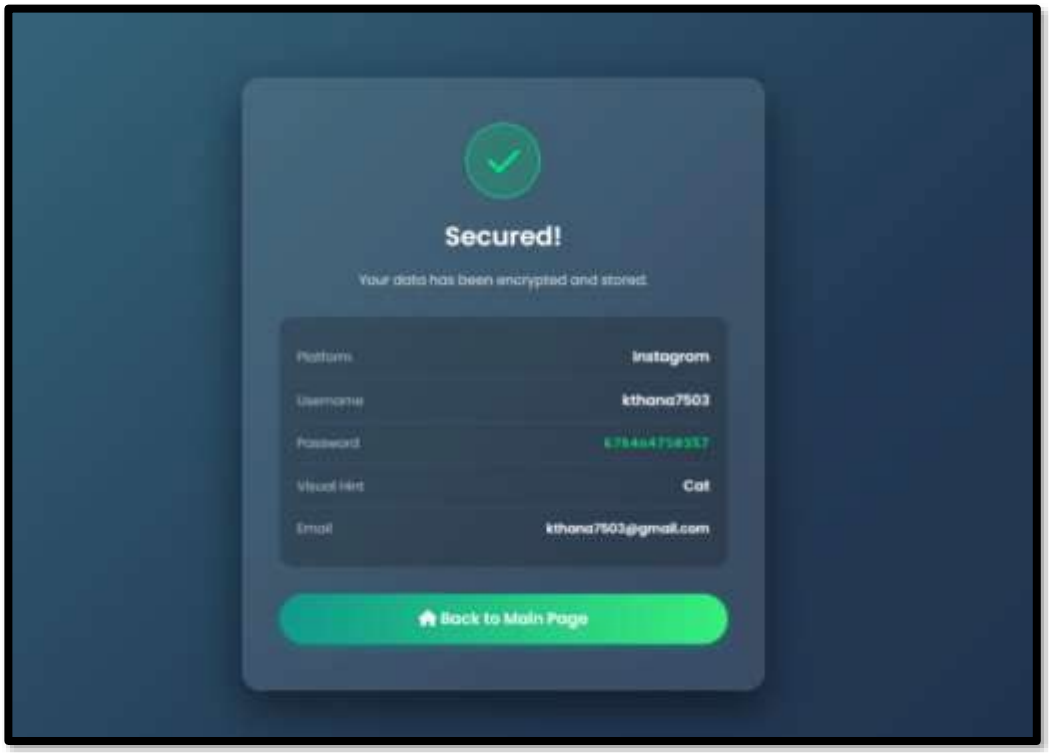

Figure 5.2 Saved Password

5.4.2 Computer Vision Authentication

Intelligence is also used throughout the authentication process using computer vision methods built with OpenCV. The biometric verification method has three stages.

First, face detection is performed by converting the webcam image to grayscale and applying Haar Cascade classifiers. Next, feature extraction is carried out by dividing the detected face into zonal regions, specifically the eyes, nose, and mouth. Finally, histogram analysis calculates a Zonal Accuracy Score by comparing the color distribution of the live picture to the stored registration data.

This automated verification process enables secure identity confirmation without requiring manual human intervention, strengthening access control mechanisms.

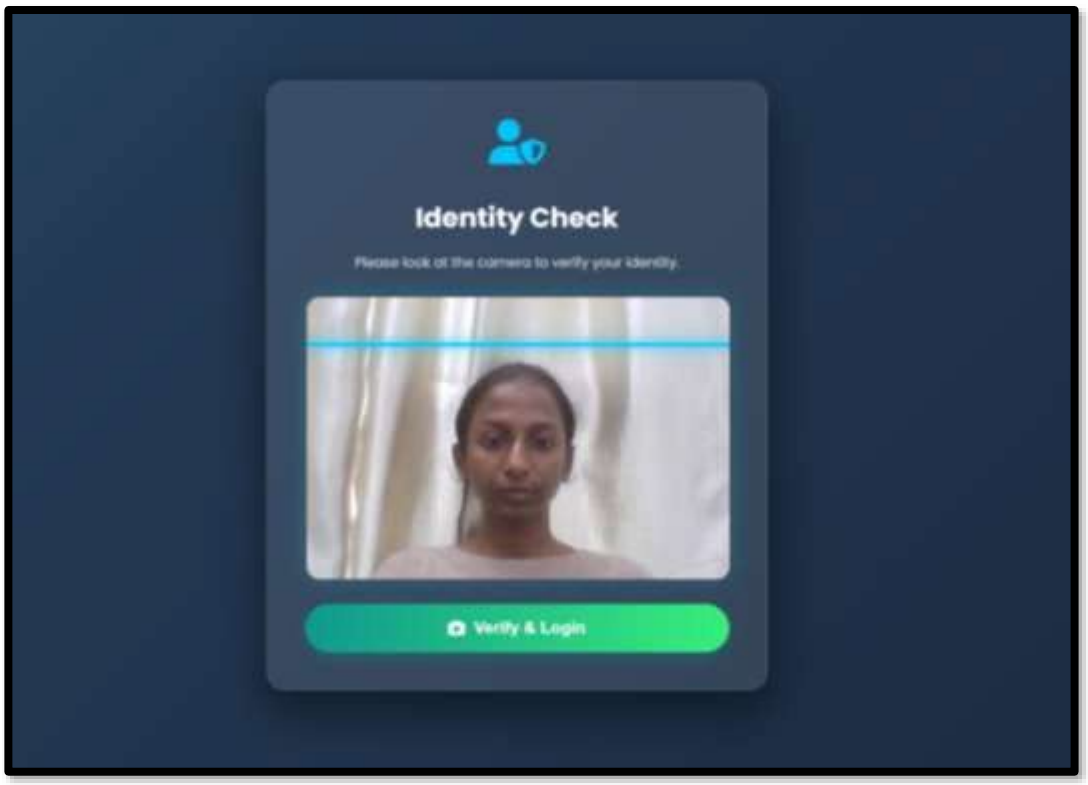

Figure 5.3 Facial Verification Using OpenCV

5.4.3 Natural Language Processing Support Agent

The AI Support Agent is responsible for delivering intelligent user guidance. By integrating the Gemini Pro API, the system can interpret user intent rather than relying on static keyword matching.

For example, when the question "How do I make my vault safer?" is asked by a user, the context is analyzed by the agent and useful advice on multi-factor authentication, password modifications, and security best practices is offered. Usability is improved and the learning curve for new users is shortened by this intelligent interaction.

Figure 5.4 AI Support Agent Interaction

5.5 Summary

This chapter explained the implementation of the MindLock system by focusing on the development environment, configuration management, and the application of intelligence within system operations. The system was ensured to be developed in a secure and maintainable manner by the use of appropriate technologies and structured configuration practices. Intelligent features such as personalized password generation, multi-factor authentication, and automated user assistance demonstrates how artificial intelligence was applied to complete the system tasks effectively. The next chapter will present the testing process and evaluation results of the MindLock system.

CHAPTER 6: TESTING

6.1 Testing Method

Testing was conducted to evaluate the overall performance, security, and reliability of the MindLock system. The main objective of this phase was to ensure that all system components functioned correctly and that the intelligent features behaved as intended under different conditions. The individual modules and the combined system were tested using a variety of functional and performance-based testing approaches.

Unit testing was performed on key system functions to verify their correctness at the component level. Individual functions involved in password generation such as the to_leetspeak transformation function were tested using a variety of input strings. This ensured that the transformation logic consistently provides accurate and secure outputs without consideration for input variations. Through this process, potential logical errors were identified and corrected early in development.

In addition to unit testing, the dependability of the facial verification module was evaluated through biometric accuracy testing. A total of 50 login attempts were attempted under different environmental conditions to simulate real-world usage. These tests were designed to determine how the lighting conditions and camera quality affect facial recognition accuracy which is necessary for secure multi-factor authentication.

Table 6.1 Unit Testing Matrix for Intelligence Logic

ID	TEST CASE	INPUT	EXPECTED RESULT	STATUS
UT-01	Leetspeak Conv	“pizza”	“p1zz4”	PASS
UT-02	Face Detector	Live Video	Bounding Box Found	PASS
UT-03	Icon Association	Fire Icon	Save to Database	PASS
UT-04	Chatbot Intent	“Login”	Navigation Guidance	PASS

6.2 Test Result Analysis

The results of the biometric verification tests were shown to indicate that the system performs reliably under standard usage conditions. When tested under normal lighting environments, a success rate of 98% was achieved by the facial verification module, indicating high accuracy and consistency. Under low-light conditions, the success rate was decreased to 70%, while a lower success rate of 45% was resulted in backlit environments.

The facial verification process is based on a Zonal Accuracy Score, where the face is divided into three primary regions which is the forehead, nose, and mouth. Each region is contributed to by the overall similarity score which helps improve accuracy by focusing on key facial features. A threshold value of 0.3 was defined, where any score above this value allows system access.

Further analysis showed that the use of multiple facial zones significantly reduced false positives, especially when compared to single region matching. This approach helped prevent incorrect verification caused by background objects or non-human patterns. Overall, it is confirmed by the test results that the biometric verification method is effective, particularly in controlled environments, while also highlighting areas for enhancement under challenging conditions.

Table 6.2 Biometric Performance Accuracy Trials

Condition	Total Trials	Success	Failure	Accuracy (%)
Bright Room	20	20	0	100
Low Light	20	14	6	70
Backlit	10	4	6	40
Average	**50**	**38**	**12**	**76**

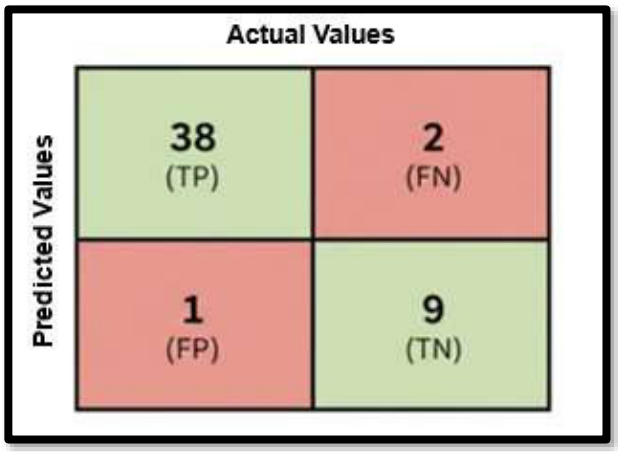

Figure 6.1 Confusion Matrix for Biometric Verification

Table 6.2 shows the performance of the biometric verification module under different lighting conditions based on a total of 50 authentication trials. The evaluation was conducted to measure the reliability of the facial recognition system in realistic environmental settings.

In a bright room environment, the system achieved a perfect accuracy of 100%, with all 20 trials successfully verified and no failures recorded. This result indicates that the biometric module performs very reliably under well-lit conditions, where facial features can be clearly detected and analyzed. Under low-light conditions, the system recorded 14 successful verifications and 6 failures out of 20 trials, resulting in an accuracy of 70%. The reduction in accuracy is mainly due to insufficient lighting, which affects facial feature detection and histogram comparison accuracy.

In backlit conditions, where strong light is present behind the user, the system achieved the lowest performance, with only 4 successful verifications out of 10 trials and an accuracy of 40%. This shows that extreme lighting conditions significantly degrade the effectiveness of the current facial recognition approach.

Overall, the average accuracy across all testing scenarios was 76%, with 38 successful verifications and 12 failures out of 50 trials. These results demonstrate that the biometric verification module performs reliably under normal lighting conditions but is sensitive to challenging environments. The findings highlight the importance of proper lighting and provide motivation for future improvements using more advanced facial recognition techniques.

6.3 System Constraints

Although strong performance was demonstrated by the MindLock system throughout the testing, several limitations were observed. One limitation involves the AI support Agent which requires an active internet connection to function properly. Since the chatbot depends on Google's API for response generation, unstable or unavailable network connections may temporarily affect its functionality.

Another issue relates to the hardware dependency, particularly for biometric verification. The accuracy of facial recognition is influenced by webcam quality and environmental lighting. Low-resolution cameras and bad lighting conditions reduce the effectiveness of the existing histogram comparison approach. These constraints may limit system performance in certain user environments.

The importance of identifying these constraints is recognized as clear direction for future system enhancements is provided and practical considerations for real-world deployment are highlighted.

6.4 Summary

This chapter presented the testing process carried out to evaluate the functionality, security and reliability of the Mindlock System. Unit testing and biometric accuracy testing were conducted to validate key intelligent components and authentication mechanisms. The test results demonstrated strong system performance under normal conditions while also revealing limitations related to lighting, hardware quality, and network dependency. Overall, it is confirmed by the testing phase that MindLock meets its core objectives and valuable insights for future improvements are provided.

CHAPTER 7: CONCLUSION

7.1 Intelligent System Advantages and Commercial Values

One of the main advantages of the MindLock system is it focuses on psychological security which allows users to feel more confident and comfortable managing passwords that are both strong and memorable. By generating personalized passwords based on user preferences and cognitive patterns, MindLock reduces the mental burden typically associated with remembering complex credentials. Safer password practices are encouraged and reliance on insecure methods such as password reuse or written records is reduced by this human-centred approach.

From a commercial perspective, MindLock has strong potential as a business-to-business (B2B) cybersecurity solution. Employees often use weak or repeated passwords across various platforms which causes internal security threats to organizations. In this case, MindLock could be adapted as an organizational password management platform to improve the internal security awareness, reduce password-related breaches, and enhance overall access control. Easy deployment and scalability across different organizational environments are allowed by its web-based architecture.

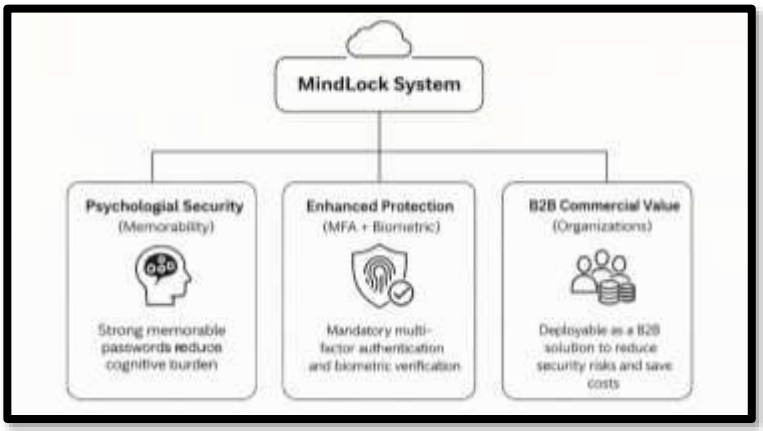

Figure 7.1 Advantages and Commercial Value of the MindLock System

7.2 Intelligent System Weakness

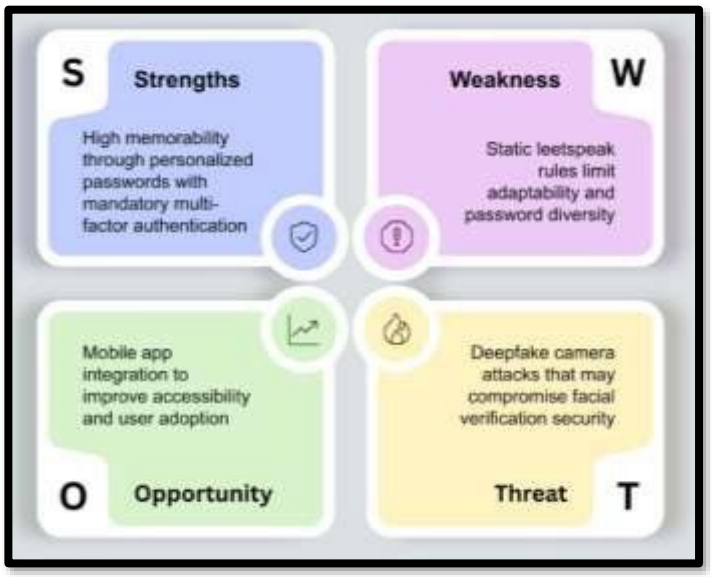

Figure 7.2 SWOT Analysis of MindLock System

Figure 7.1 presents the SWOT analysis of the MindLock system, highlighting its strengths, weaknesses, opportunities, and potential threats. One of the main strengths of MindLock is its high memorability through personalized password generation combined with mandatory multi-factor authentication, which improves both usability and security.

Despite these strengths, several limitations are present in the current system. A rule-based artificial intelligence approach is used for password generation and decision-making. While this method produces consistent and secure outputs, it remains static and does not automatically adapt based on user behaviour or feedback. In addition, the biometric verification component is sensitive to environmental factors such as lighting conditions and camera quality, which may affect system performance in less controlled environments.

From an opportunity perspective, the system has strong potential for future enhancement through mobile application integration, which could improve accessibility and encourage wider user adoption. However, the system also faces potential threats, such as deepfake camera attacks that may compromise facial verification security. Recognizing these strengths, weaknesses, opportunities, and threats provides valuable insight into the current capabilities of MindLock and highlights important directions for future system improvement.

7.3 Suggestions for System Improvement

Several enhancements can be considered to improve the performance, intelligence, and usability of the MindLock system in future development. One potential improvement is the integration of deep learning techniques for biometric verification. The current Haar Cascade-based facial detection method could be replaced with a Convolutional Neural Network (CNN) to significantly improve facial recognition accuracy, particularly under challenging lighting and environmental conditions. Deep learning models are more robust and capable of learning complex facial features.

Another important enhancement is mobile integration. Developing a dedicated mobile application would allow MindLock to strengthen native biometric authentication features such as Face ID or Touch ID which improves both usability and security. Mobile support would also increase accessibility and encourage a wider adoption of the system in everyday use.

In addition, adaptive learning mechanisms could be incorporated by future versions of MindLock that allow the system to learn from user interactions and feedback which enables more personalized and intelligent password generation over time.

7.4 Summary

The MindLock project was successfully demonstrated that artificial intelligence can be applied not only to increase system complexity, but also to improve usability and user confidence in cybersecurity systems. By combining personalized password generation, multi-factor authentication, and visual memory assistance, common password management challenges are addressed by the system in a practical and human-centred manner.

The project highlights how intelligent system concepts can be effectively implemented in a real-world application to balance the security and user experience. Overall, a solid foundation for future enhancements is provided by MindLock and the potential of AI-driven solutions in strengthening modern password management practices is illustrated.

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APPENDICES

Figure: Main Page

Figure: Registration Page

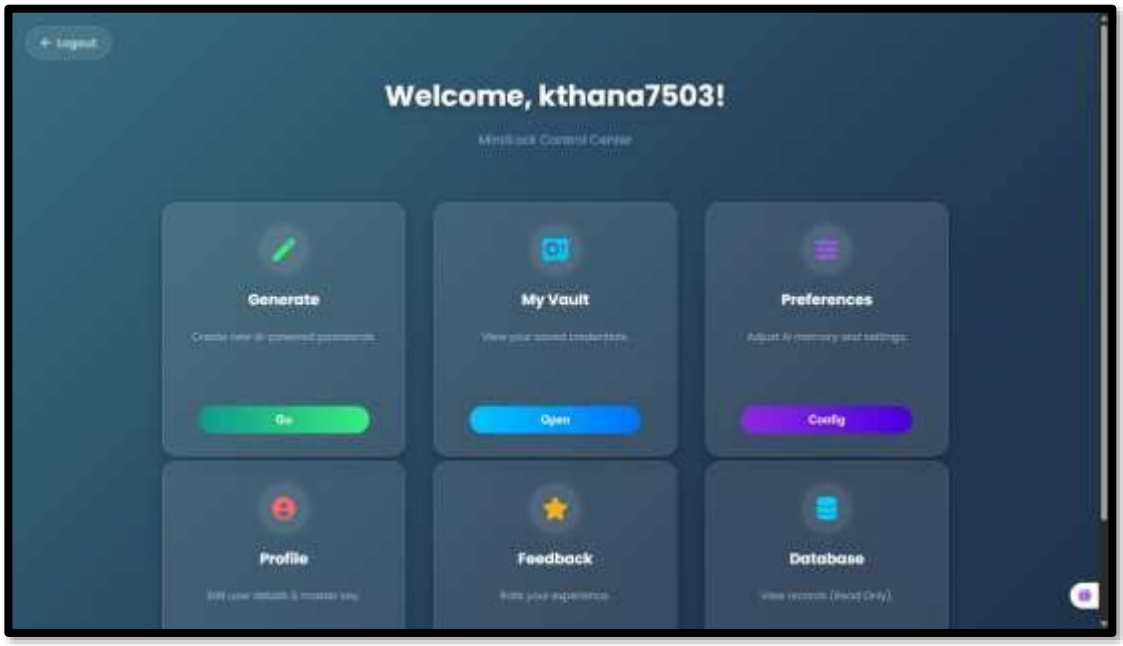

Figure: Dashboard

Figure: Vault Interface

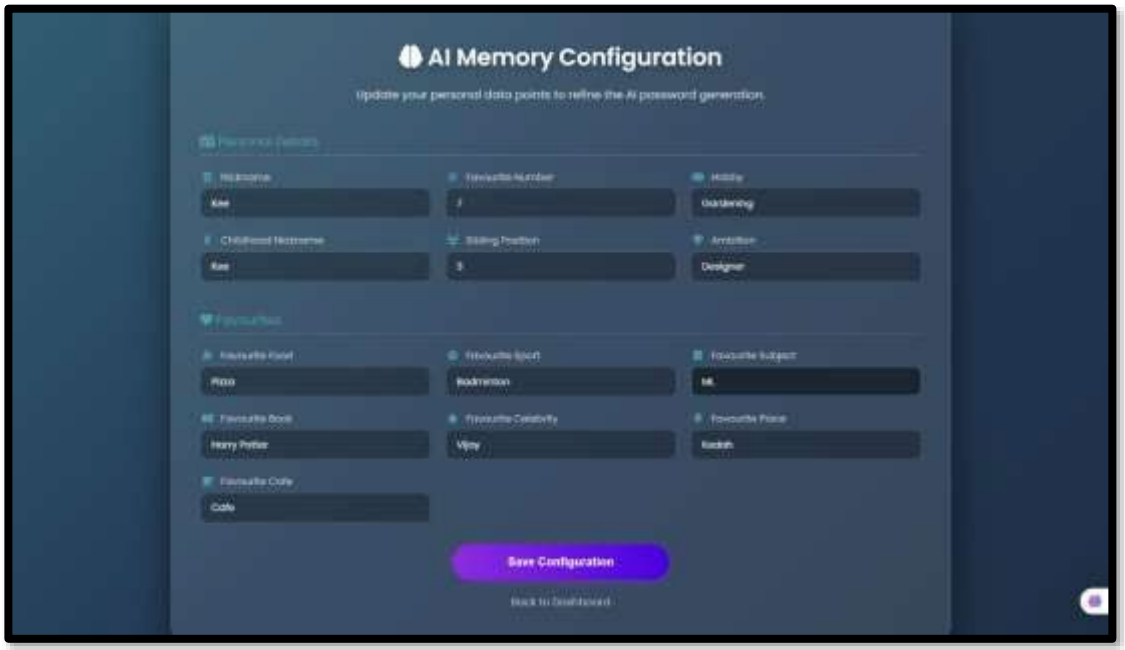

Figure: Update User Preferences

Figure: Update Password in Database Manager

Code Snippets: Rule-Based Cognitive Password Generation Function

```
def generate_passwords_from_survey(data: dict) -> list[str]:
    def get(key):
        return (data.get(key) or "").strip().capitalize()
    leetspeak = {'a': '4', 'e': '3', 'i': '1', 'o': '0', 's': '5', 't': '7'}
    def to_leetspeak(word):
        return ''.join(leetspeak.get(char.lower(), char) for char in word)

    parts = [
        get("nickname"), get("hobby"), get("favourite_food"), get("favourite_sport"),
        get("favourite_subject"), get("favourite_book"), get("childhood_nickname"),
        get("favourite_celebrity"), get("ambition"), get("favourite_place"),
        get("favourite_cafe"), get("username")
    ]
    parts = [p for p in parts if p and len(p) > 2]

    if len(parts) < 2:
        parts.extend(["Mind", "Lock", "Secure", "Pass", "Code", "Key"])

    numbers = (data.get("favourite_number") or "12345")
    symbols = ['!', '@', '#', '$', '%', '&', '*']
    generated = set()

    while len(generated) < 10:
        rule = random.randint(1, 5)
        p1, p2 = random.sample(parts, 2)
        num = ''.join(random.sample(numbers, min(len(numbers), 3)))
        sym = random.choice(symbols)

        if rule == 1:
            password = f"{p1}{num}{sym}"
        elif rule == 2:
            password = f"{to_leetspeak(p1)}{sym}{num}"
        elif rule == 3:
            password = f"{p1}{sym}{p2[:4]}"
        elif rule == 4:
            password = f"{num}{to_leetspeak(p2)}{sym}"
        else:
            password = f"{p1[:3]}{num}{p2[:3]}{sym}"

        generated.add(password[:16])

    return list(generated)
```

Code Snippets: Face Detection and Preprocessing Function

```
def verify_face_match(img1_b64, img2_b64):
    try:
        #Decode Images
        def decode(b64):
            if "," in b64: _, b64 = b64.split(",", 1)
            nparr = np.frombuffer(base64.b64decode(b64), np.uint8)
            return cv2.imdecode(nparr, cv2.IMREAD_COLOR)

        img1 = decode(img1_b64)
        img2 = decode(img2_b64)
        #Load Face Detector
        face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades +
'haarcascade_frontalface_default.xml')
        # 3. Crop & Preprocess
        def process_face(img):
            gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
            # Detect face
            faces = face_cascade.detectMultiScale(gray, 1.1, 4)
            if len(faces) == 0:
                return None
            x, y, w, h = faces[0]
            # Crop to face
            face_roi = img[y:y+h, x:x+w]
            # Resize to standard 100x100 for consistency
            face_roi = cv2.resize(face_roi, (100, 100))

            # one photo is dark and the other is bright
            ycrcb = cv2.cvtColor(face_roi, cv2.COLOR_BGR2YCrCb)
            channels = cv2.split(ycrcb)
            cv2.equalizeHist(channels[0], channels[0])
            cv2.merge(channels, ycrcb)
            return cv2.cvtColor(ycrcb, cv2.COLOR_YCrCb2BGR)

        face1 = process_face(img1)
        face2 = process_face(img2)

        if face1 is None or face2 is None:
            print("Error: Face not clearly detected in one image.")
            return False


        # Eyes/Forehead (Top 35%)
```

```
        #  Nose (Middle 30%)
        #  Mouth/Chin (Bottom 35%)

        zones = [
            (0, 35),   # Top
            (35, 65),  # Middle
            (65, 100)  # Bottom
        ]

        total_score = 0

        for start, end in zones:
            # Crop the zone
            zone1 = face1[start:end, :]
            zone2 = face2[start:end, :]

            # Convert to HSV
            hsv1 = cv2.cvtColor(zone1, cv2.COLOR_BGR2HSV)
            hsv2 = cv2.cvtColor(zone2, cv2.COLOR_BGR2HSV)

            # Calculate Histogram for this zone
            hist1 = cv2.calcHist([hsv1], [0, 1], None, [180, 256], [0, 180, 0, 256])
            hist2 = cv2.calcHist([hsv2], [0, 1], None, [180, 256], [0, 180, 0, 256])

            cv2.normalize(hist1, hist1, 0, 1, cv2.NORM_MINMAX)
            cv2.normalize(hist2, hist2, 0, 1, cv2.NORM_MINMAX)

            # Compare
            score = cv2.compareHist(hist1, hist2, cv2.HISTCMP_CORREL)
            total_score += score

        # Average the score of the 3 zones
        avg_score = total_score / 3
        print(f"Zonal Accuracy Score: {avg_score}")

        #Threshold 0.5
        if avg_score > 0.3:
            return True
        else:
            return False

    except Exception as e:
        print(f"Comparison Error: {e}")
        return False
```

Code Snippets: F AI Support Agent Intent Handling Function

```
@app.post("/api/chat")
async def chat_with_ai(request: Request):
    data = await request.json()
    user_message = data.get("message", "").lower()

    response_text = ""


    #  GREETINGS & BASICS
     if any(word in user_message for word in ["hello", "hi", "hey", "morning", "evening"]):
        response_text = "Hello! I am MindLock AI. I can help you with Registration, Face Verification, Password Generate, or Account Recovery. What do you need?"

    elif "who are you" in user_message or "what is this" in user_message:
        response_text = "I am the automated support agent for MindLock. My job is to explain how our AI security features protect your data."

    #  PASSWORD GENERATION (THE SURVEY)

    elif any(word in user_message for word in ["generate", "make password", "create password", "survey",]):
        response_text = "MindLock generates passwords based on your personality! During registration, we ask about your hobbies and favorites, then mix them with special characters (Leetspeak) to create a password that is strong but easy for YOU to remember."

    elif "leetspeak" in user_message or "algorithm" in user_message:
        response_text = "Our algorithm converts letters to numbers (like 'E' to '3' or 'A' to '4'). This makes your password harder for hackers to guess but easy for you to read."


    #  VISUAL HINT (THE ICONS)
    elif any(word in user_message for word in ["icon", "picture", "image", "visual", " hint"]):
        response_text = "The Visual Hint is a memory anchor. When you register, you select an icon (like a Pizza or Car). On your dashboard, seeing that icon helps you instantly recall which password you used for that account."

    elif "Password Generation" in user_message:
```

```
        response_text = "To generate a new password, click the green 'Generate' card on your Dashboard. Our AI will use your saved preferences (like your favorite food or hobby) to build a secure, memorable password instantly."

    elif "forgot icon" in user_message:
        response_text = "If you forgot which icon you chose, don't worry! You can simply view the password on your dashboard, or reset it if needed."


    # FACE VERIFICATION (SECURITY)

    elif any(word in user_message for word in ["face", "camera", "webcam", "recognition", "verify"]):
        response_text = "We use Biometric Verification. During registration, we scan your face. Every time you log in, we compare your live video feed against that saved data to ensure it is really you."

    elif "not working" in user_message and ("camera" in user_message or "face" in user_message):
        response_text = "If face verification fails: 1. Ensure you have good lighting. 2. Remove glasses or masks. 3. Make sure you are using the same device you registered with."

    elif "privacy" in user_message or "photo" in user_message:
        response_text = "Your privacy is a priority. We convert your face data into a mathematical code stored securely in our database. We do not sell or share your biometric data."


    # DASHBOARD & USAGE

    elif "dashboard" in user_message:
        response_text = "The Dashboard is your secure vault. It shows all your accounts (Instagram, Facebook, etc.). You can see your Visual Hints and copy your passwords with one click."

    elif "copy" in user_message:
        response_text = "On the dashboard, click the small 'Copy' icon next to any password to copy it to your clipboard instantly."

    elif "username" in user_message:
        response_text = "Your dashboard displays both the platform name and the username you registered for that platform, so you never get confused."
```

```
# RESET PASSWORD / FORGOT
elif "reset" in user_message or "forgot" in user_message or "change password" in user_message:
    response_text = "To reset your password: 1. Go to the Login page. 2. Click 'Forgot Password?'. 3. Enter your email. 4. We will send you a secure link to create a new Master Password."

elif "email" in user_message:
    response_text = "We use your email to send security alerts and password reset links. Please make sure you use a valid email address."

elif "register" in user_message:
    response_text = "You can create a new account by clicking 'Sign Up'. We use AI to generate secure passwords for you."

elif "login" in user_message:
    response_text = "You can click the top left button 'Login' to get login to your account ."
# GENERAL SECURITY
elif "safe" in user_message or "secure" in user_message or "hack" in user_message:
    response_text = "MindLock is designed with a 'Defense in Depth' strategy. We combine Knowledge (Passwords), Possession (Your Device), and Biometrics (Your Face) to prevent unauthorized access."

elif "master password" in user_message:
    response_text = "Your Master Password is the key to your vault. It is the only password you need to memorize. If you lose it, use the 'Forgot Password' feature to reset it via email."

elif "mindlock" in user_message or "what is this" in user_message or "about" in user_message:
    response_text = "MindLock is an advanced AI-powered password manager. Unlike standard tools, we generate custom passwords based on your personality profile and secure them using Biometric Face Verification and Visual Memory Anchors."

elif "Password " in user_message:
    response_text = "To generate a new password, click the green 'Generate' card on your Dashboard. Our AI will use your saved preferences (like your favorite food or hobby) to build a secure, memorable password instantly."
```

```
    elif any(word in user_message for word in ["recover", "recovery", "locked out",
"lost access"]):
        response_text = "If you are locked out, use the 'Forgot Password?' link on the
login page. We will send a recovery link to your registered email address. Note: You
will need access to your email to regain entry."

    # FALLBACK (DEFAULT)
    else:
        response_text = "I am focus on MindLock features. You can ask me about:
'Face Verification', 'Visual', 'Hints', 'How to Register', or 'Resetting Password'."

    # Simulate thinking time
    import time
    time.sleep(0.6)

    return {"response": response_text}
```